SA 220

SITE SELECTION FOR POND CONSTRUCTION

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SITE SELECTION FOR POND CONSTRUCTION

- At the end of this lecture, students should:

1. understand the criteria for choosing good sites for fish ponds
2. understand the best sites.
3. Know how to prepare a site for pond
4. know how a fish pond is constructed

SITE SELECTION

- Site selection in aquaculture (i.e. fish farming) is the selection of an ideal place for setting up an aquaculture facility
- A properly designed and constructed pond will be easily managed and will last longer, saving extra work and maximizing profit

MAIN FACTORS TO CONSIDER

1. **Water**

- Enough all year-round
- An unpolluted
- Ideally ponds must be filled and drained by gravity

MAIN FACTORS TO CONSIDER

2. Soil type

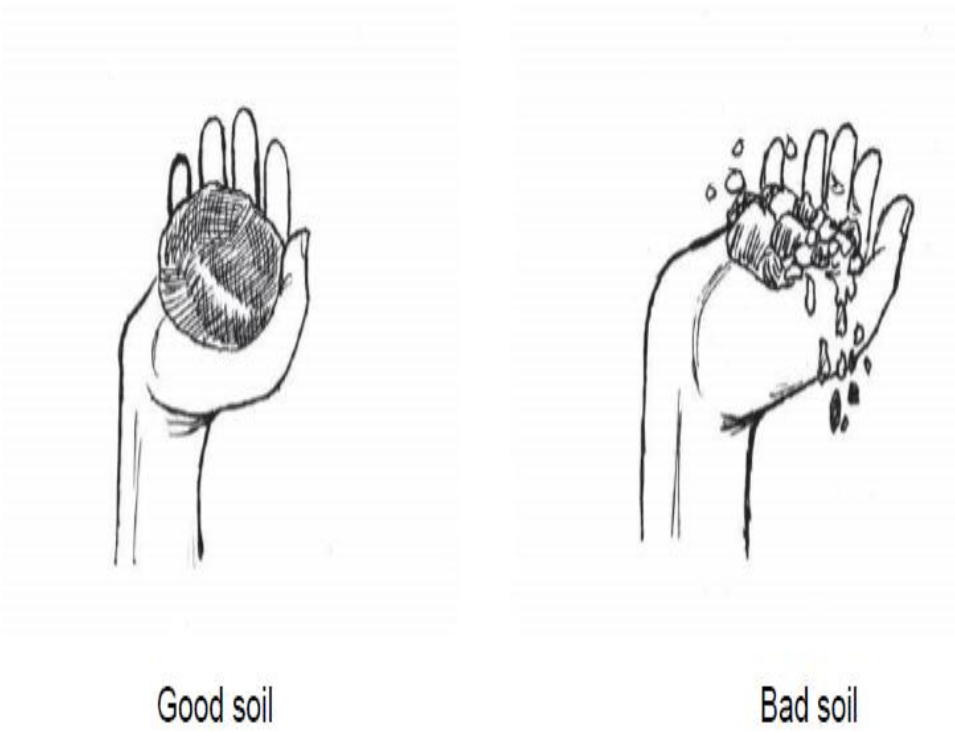
Soil containing clay is the best type for fish ponds while sandy soil is not good for fish ponds

(WHY?)

How do you know if your farm is having enough clay?

MAIN FACTORS TO CONSIDER

- Firstly, dig one meter deep, and take a sample from the bottom of the hole.
- Then squeeze some moist oil into a ball and toss it into the air and catch it. If the ball remains together, the soil is good, but if it falls apart, it is not suitable for building fish ponds.



Good soil

Bad soil

MAIN FACTORS TO CONSIDER

Secondly, you can also fill the hole with water and leave it overnight to confirm the water retention capacity.

3. Topography-Location

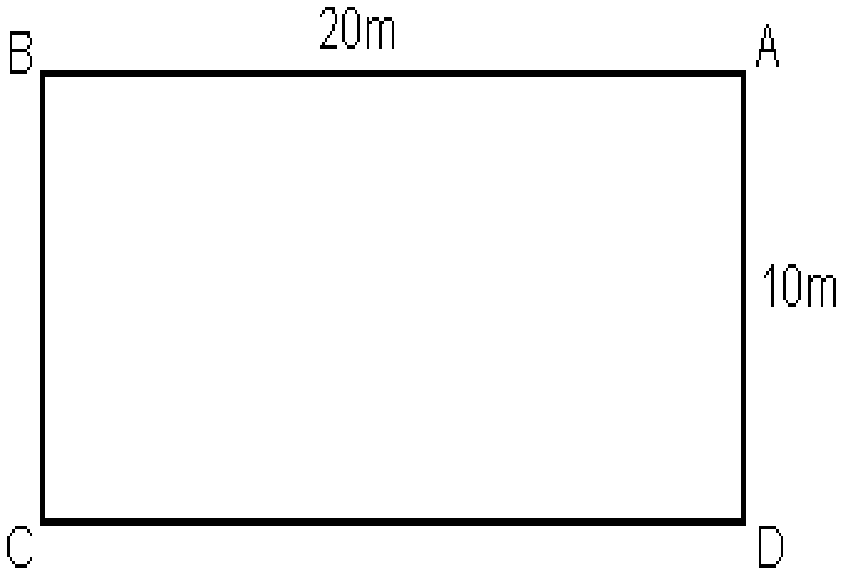
- Gently sloping land is ideal for fish farming.
- Topography will also determine the layout of the ponds

POND CONSTRUCTION.

- Once you are satisfied with the site and how the ponds will be orientated, the follow is to be done:

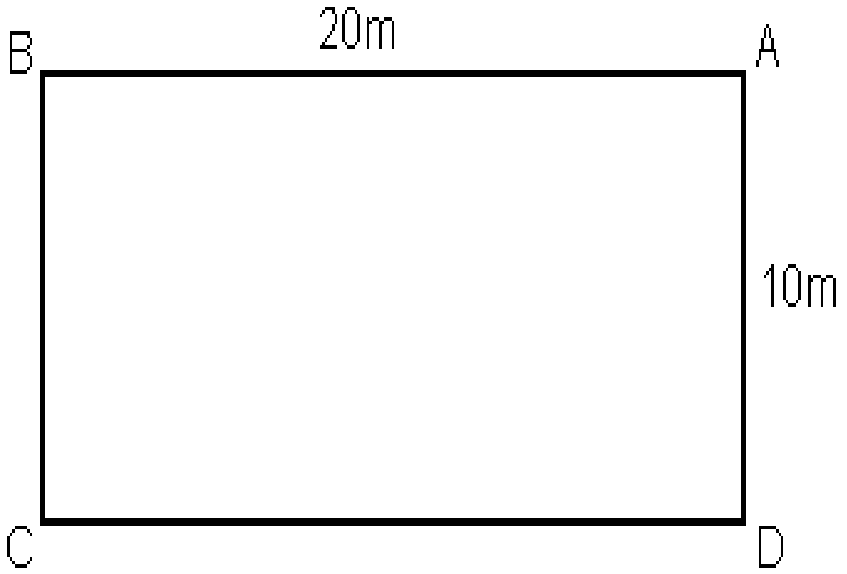
1. Clear the site; remove brush and top soil

2. Stake out the pond with sticks to indicate the dikes.



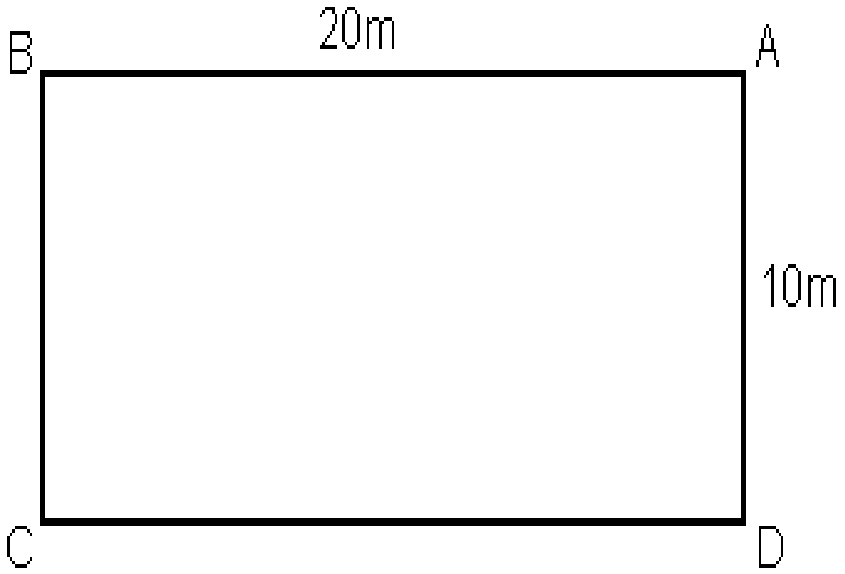
POND CONSTRUCTION.





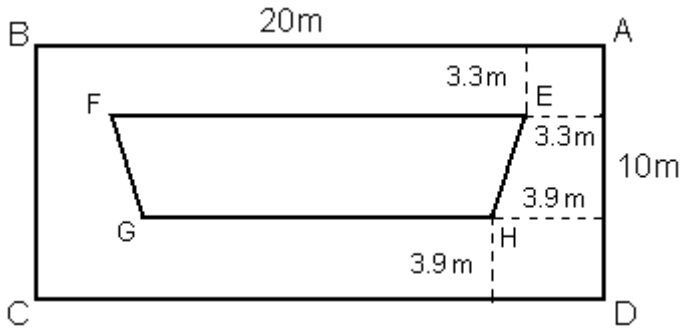
POND CONSTRUCTION.



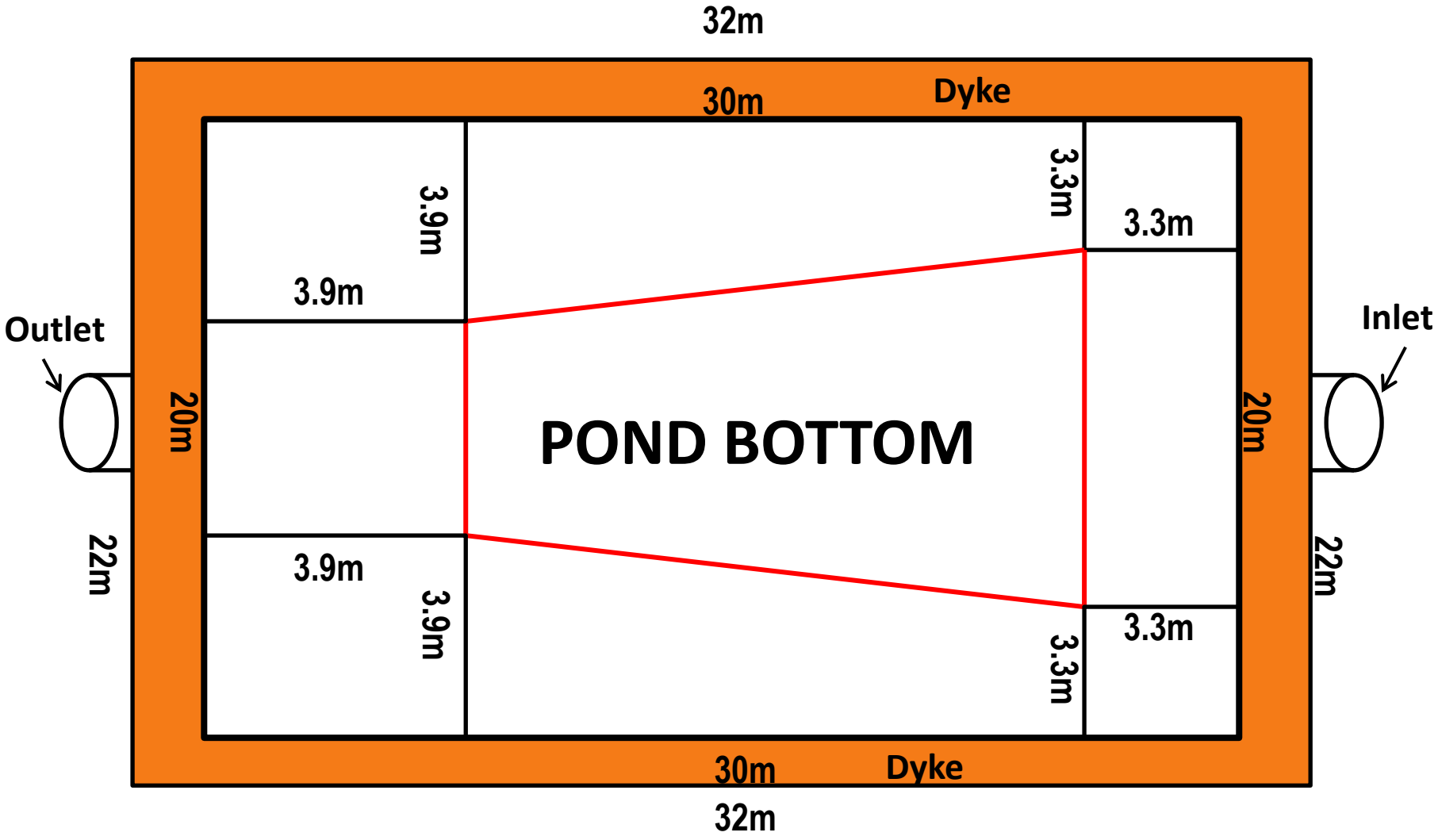


POND CONSTRUCTION.

3. Measure the pond bottom



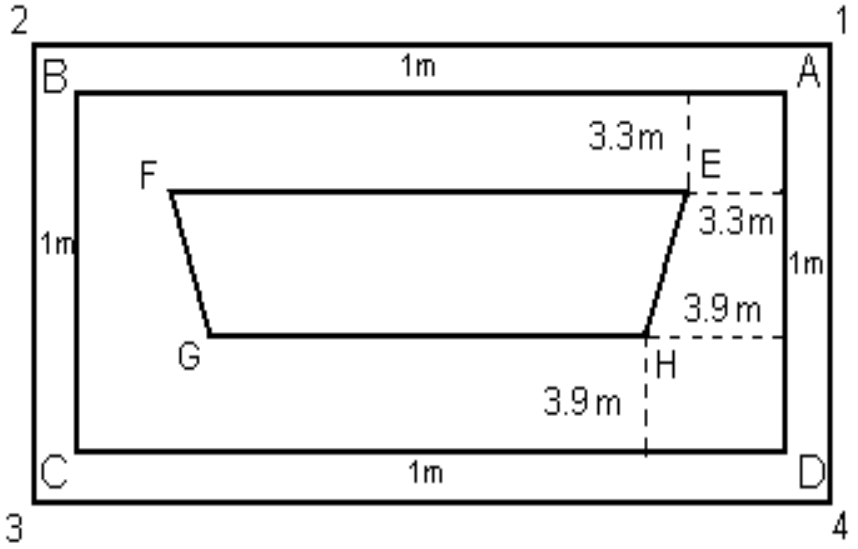
- To do this, establish the shallow and deeper end (E.g A and B is the shallow end while D and C are indicating the deeper end of the pond)
- Since the ideal depth on the shallow end is 1.1m and deeper end 1.3m, you will multiply this by 3 because the slopes inside the pond are three times than the depth.


32m
30m
Dyke
3.3m
3.3m
3.9m
3.9m
Outlet
Inlet
20m
POND BOTTOM
20m
22m
22m
3.9m
3.9m
3.3m
3.3m
30m
Dyke
32m

POND CONSTRUCTION.

4. Create the dykes

- The dike top around the pond is 1 m wide and this is the top of the pond.
- Compact the dykes so that they are strong



POND CONSTRUCTION.

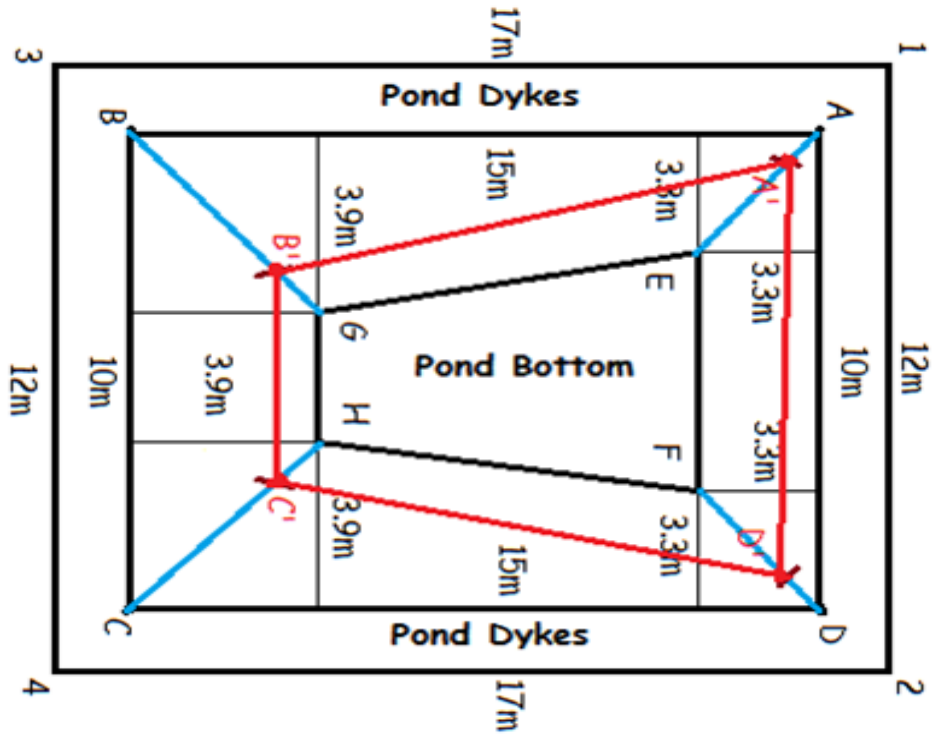
5. Cut and fill

The dike top around the pond is 1 m wide and this is the top of the pond

3. Dig the pond and build the walls

4. Compact the dikes in 15 cm thick layers of soil

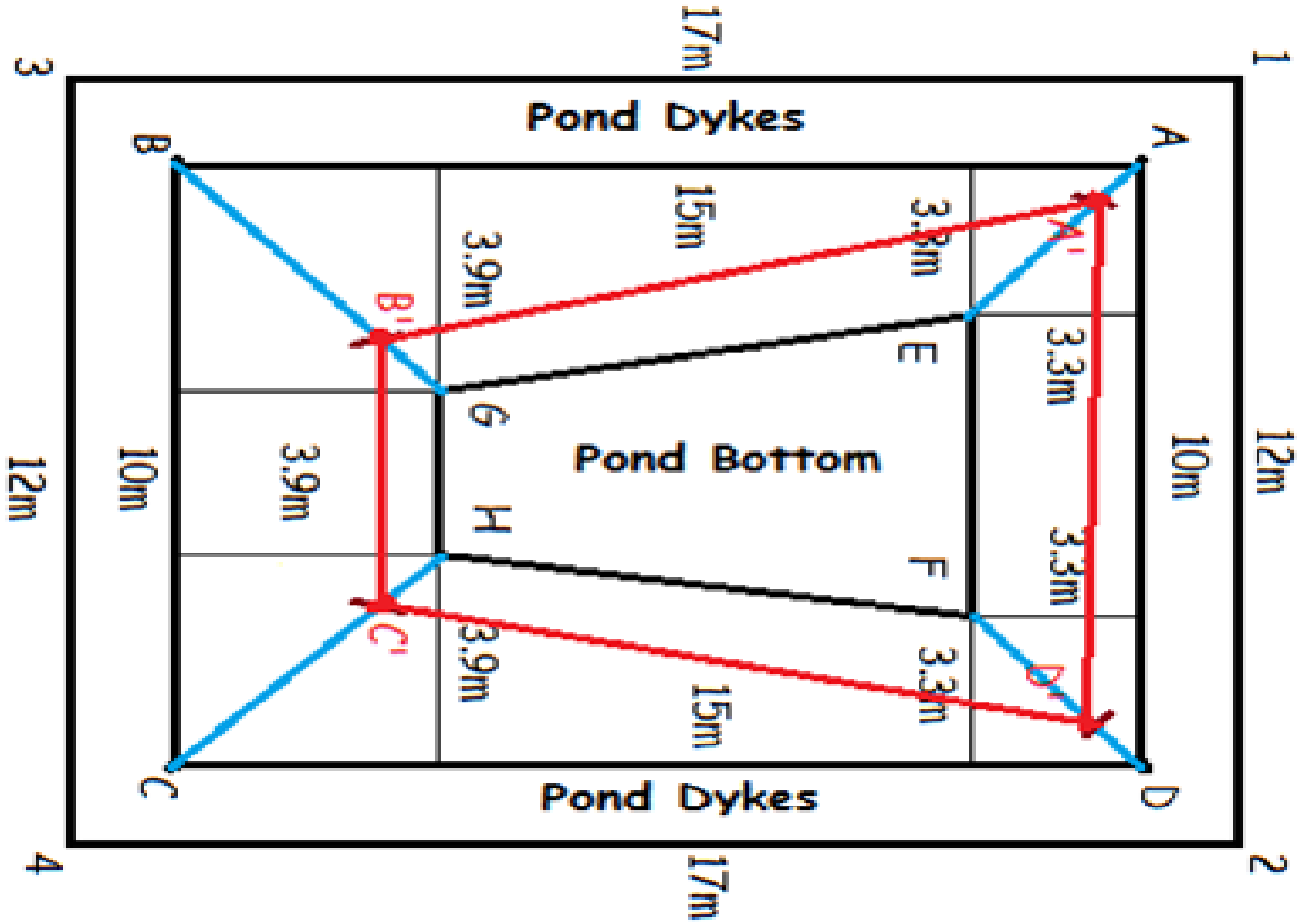
5. Put the inlet and the outlet



POND CONSTRUCTION.

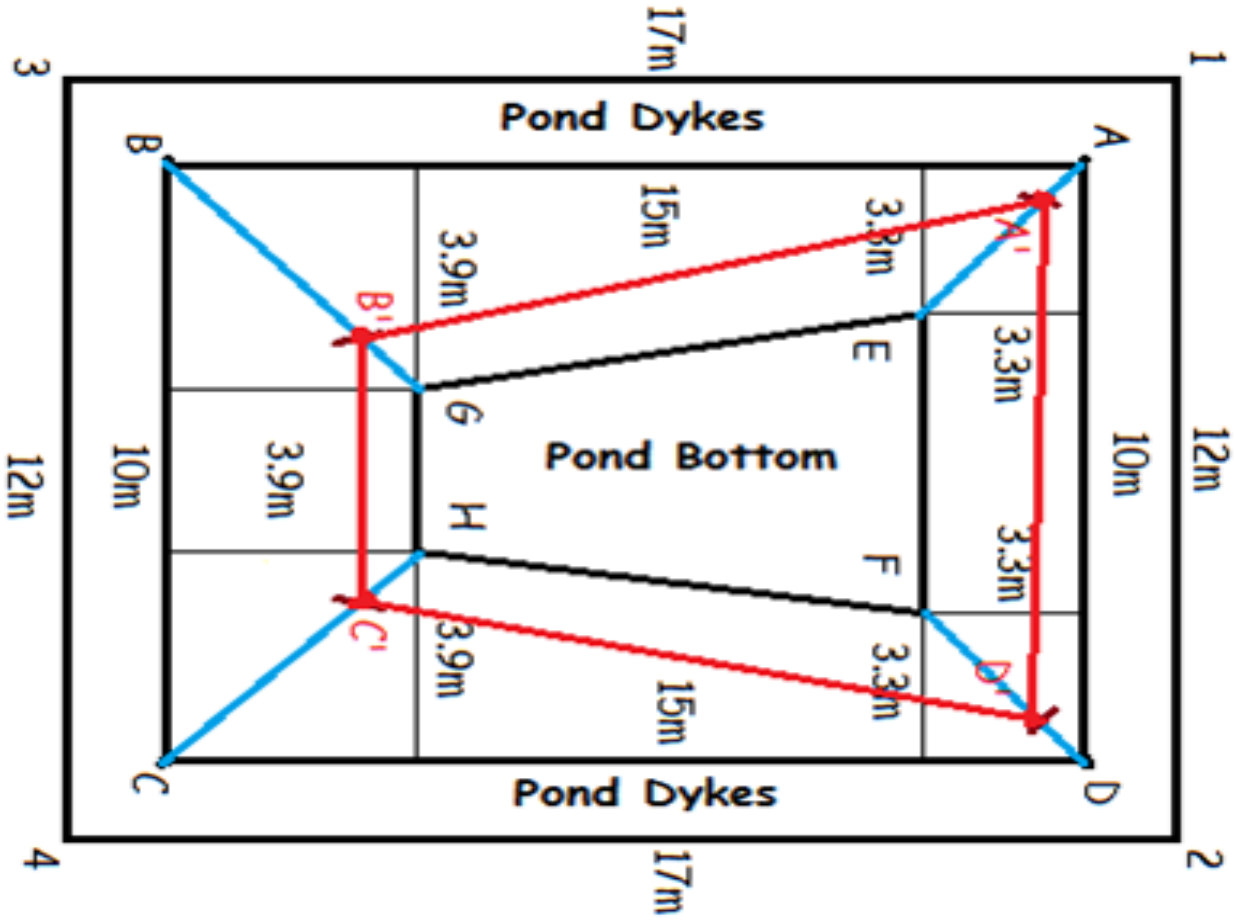
5. Cut and fill

Its not always that you will have to dig, depending on the topography of the areas, in come corners you just fill with soil will in some you just do sheaving.



POND CONSTRUCTION.

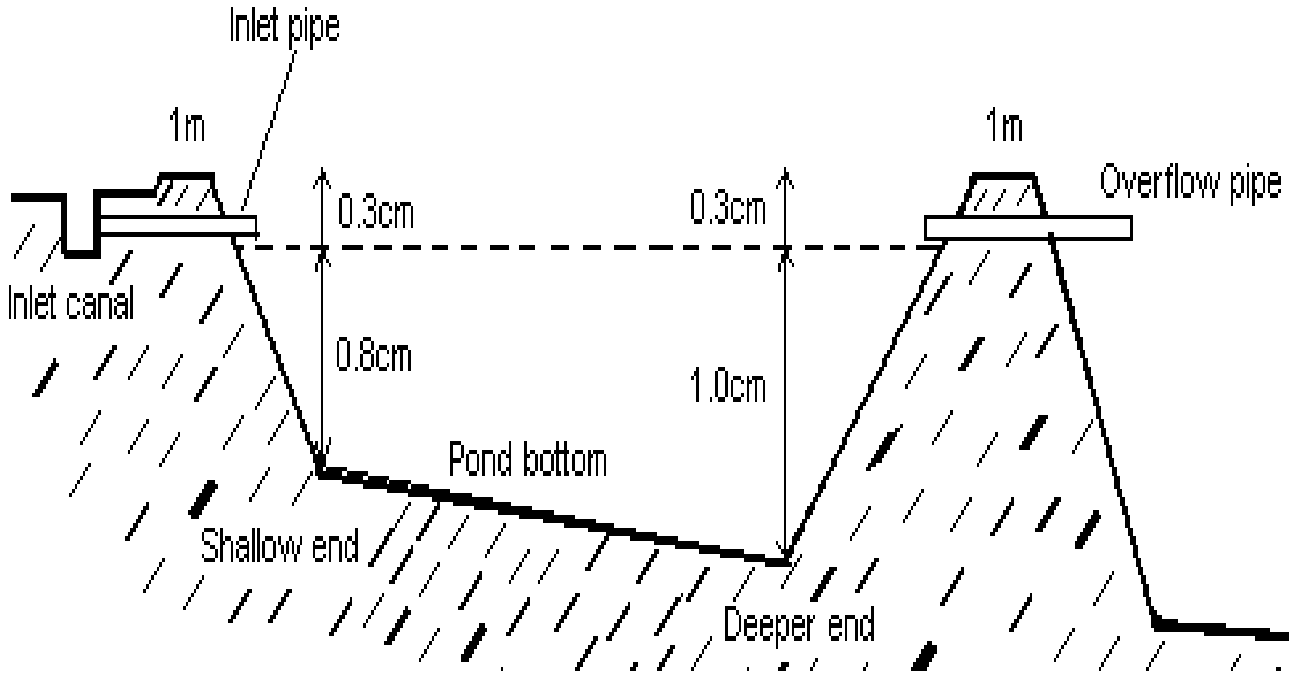


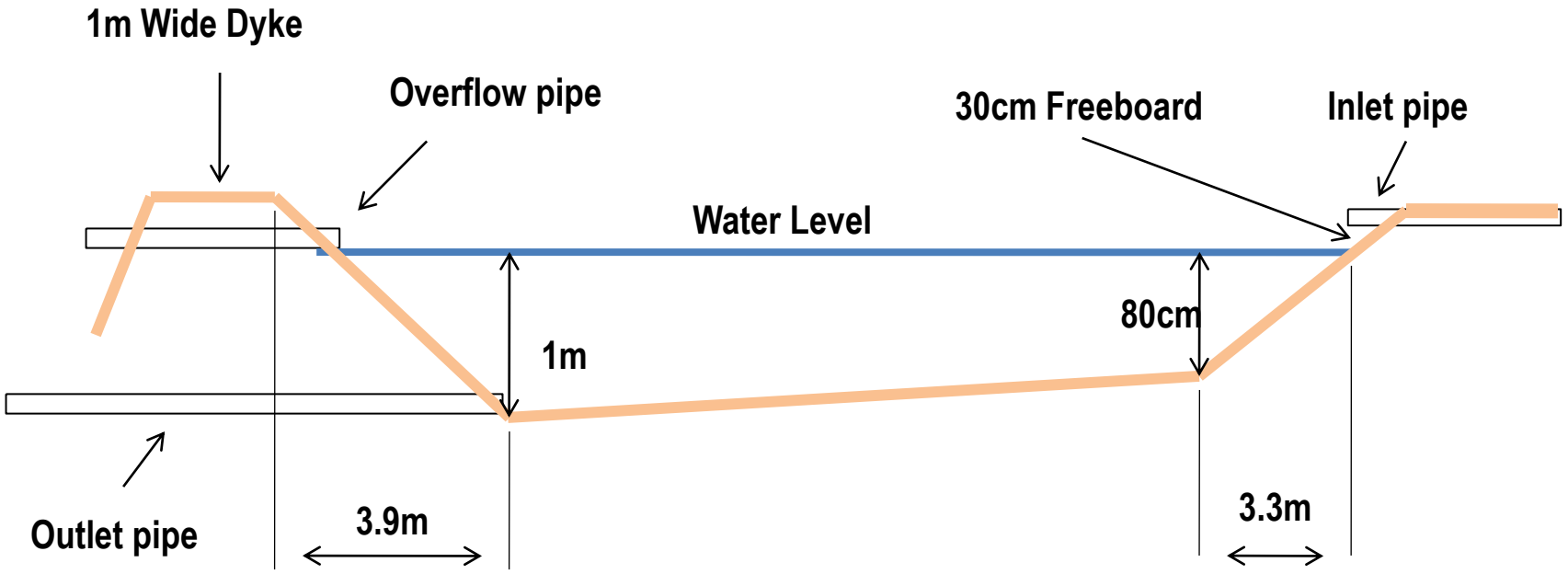


POND CONSTRUCTION.

6. Dig the pond

- Cross section of the pond of the pond is shown on the right.




1m Wide Dyke
Overflow pipe
30cm Freeboard
Inlet pipe
Water Level
80cm
1m
Outlet pipe
3.9m
3.3m

POND CONSTRUCTION.

7. Connect the water supply system

8. Fill the pond with water

9 . Check for problems before stocking fish (leaks, seepage, settling, shrinking of dikes)

POND CONSTRUCTION.

- **Recommended earthen pond standards:**

- Pond depth of 1.1m to 1.3m for ponds smaller than 1000sqm and 1.3m to 1.5m for bigger ponds
- Inner pond slopes of 1:3 ratio for smaller ponds and 1:4 ratio for bigger ponds
- Free board of 30cm for smaller ponds and 50cm for bigger ponds.

POND CONSTRUCTION.

❑ **Instruments or equipment used**

1. Measuring tape
2. String
3. Stakes (pegs)
4. Hoes
5. Peaks

POND CONSTRUCTION.

❑ **Instruments or equipment used**

6. Shovels & spades
7. Wheelbarrow
8. Compactor
9.Machinery

POND CONSTRUCTION.

- **Conclusion**

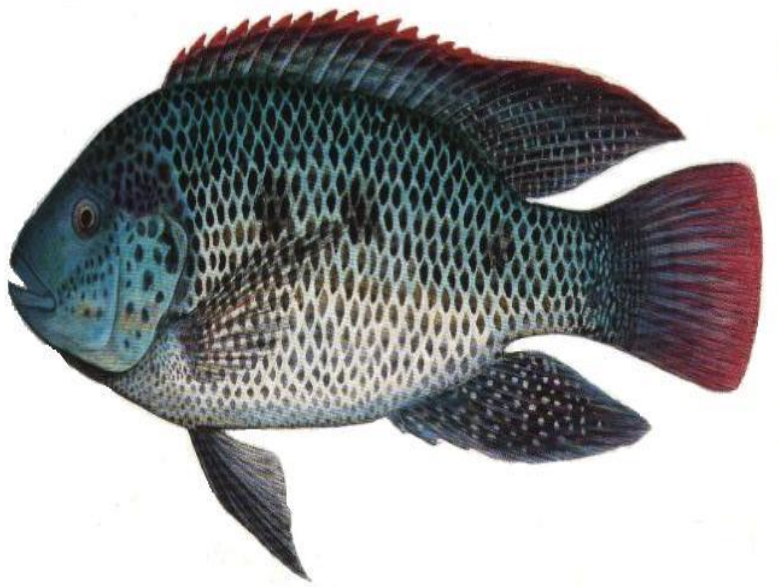
- A properly sited and constructed pond will have a direct influence on production and returns
- Consequently, poorly constructed ponds, give poorer production yields and returns;
- This is because additional management efforts and associated costs are required to achieve comparable yields

CULTURABLE SPECIES

❖ The selection of a species for culture is important and should be considered for various reasons.

- The reasons may fall in the following categories:
  - Impact on others fishes or other living organisms;
  - Social or religious concerns of the society where you want to conduct your fish culture;
  - Input requirement for its production;
  - Good growth performance;
  - Marketability.

1. Oreochromis andersonii



❖ **Common names**

- Three-spotted bream;
- Kafue bream;
- Njinji (Lozi).

❖ **External features**

- Three black spots on the body;
- Yellowish eyes;
- Several spots on the operculum (gill cover).



2. *Oreochromis macrochir*

❖**Common names and features**

- Green-headed bream;
- Long fin;
- Pale (Bemba);
- Kipale (Kaonde).



❖**External features**

- Greenish head.
- Red eyes
- General body color lighter or darker olive green.

3. *Coptodon rendalli*

❖ **Common names**

- Red breasted bream
- Impende
- Mpende



❖ **External features**

- Reddish breast.
- Bright red throat.
- Body with 5-7 dark olive vertical bars.



4. Oreochromis niloticus

❖**Common names**

- •Nile Tilapia
- •Madamu
- •Wamunyima

❖**External features**

- •Caudal fin has clear vertical stripes (zebra lines).
- •6- 7 dark vertical bars on body.
- •Dorsal fin and anterior part of caudal fin is grey



5. Oreochromis tanganicae

Common names

- Tanganyika bream;
- Mpende.



Prominent Feature

- Scales of chest and belly very small;
- Bright colour and red caudal fin



6. *Clarias gariepinus* (African catfish)

- **Common names**
  - Barbel fish;
  - Cat fish;
  - Mulamba;
  - Mulonge;
  - Mubondo.
- **External features**
  - Caudal fin is rounded;
  - Eyes small, mouth large;
  - Four pairs of long filamentous barbels.





7. Cyprinus carpio (Carp fish)



Common names

- Common carp (Full scaled carp;
- Mirror carp (Scanty scales).

External features



- Scales all over (common carp);
- Protruding mouth, with a single pair of barbels.

END.

Thank you for you attention